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DEMOGRAPHIC HISTORY OF KIEV AND PEREYASLAVL (10000BC - 1300AD)

A Regional Literature Review of Population and Settlement (10000BC - 1300AD).

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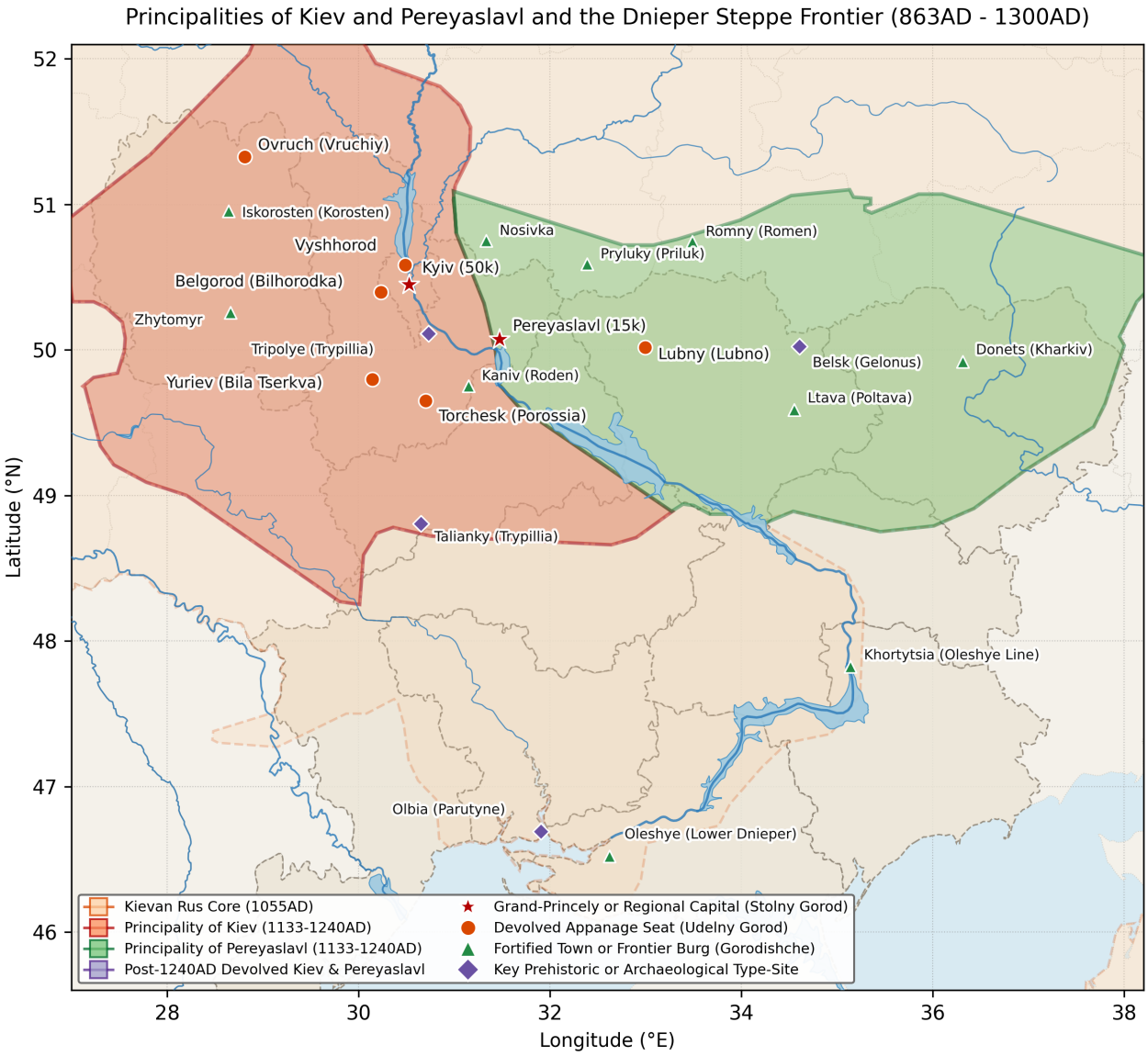


Figure 1. Regional boundaries and fortified settlements of the principalities of Kiev and Pereyaslavl across the Middle Dnieper (10000BC - 1300AD).

Abstract

This review reconstructs the population and settlement history of the Principalities of Kiev and Pereyaslavl in the Middle Dnieper Core from the Upper Palaeolithic (10000BC) to the Mongol campaigns (1239-1240AD) and the post-conquest appanage period (1250AD - 1300AD). Across the Holocene and pre-Mongol medieval era, the Middle Dnieper core experienced four main demographic cycles before the invasion of 1239-1240AD: first, the Eneolithic Trypillia BII-CI mega-site phenomenon (c. 3600BC) in the Ros-Southern Bug interfluve, when regional population reached 280k [1][2]; second, the Scythian forest-steppe hillfort and Greek Pontic colonial expansion (c. 500BC), peaking at 540k around the rampart complex of Belsk (*Gelonus*, 4.020 ha) and the Milesian polis of Olbia [3][4]; third, the Chernyakhov Culture agricultural expansion (c. 300AD), which sustained 720k inhabitants across dense open-settlement networks before the Hunnic crisis [5]; and fourth, the Kievan Rus urban and rural peak (1000AD - 1200AD), when the construction of over 950 km of timber-and-earth *Zmiyevy Valy* (Serpent's Walls) and 240 fortified *gorodishcha* (n=240) supported a regional population of 1,05M in 1000AD, 1,32M in 1100AD and 1,58M in 1200AD [6][7][8][9]. Batu Khan's storming of Pereyaslavl in March 1239AD and the siege of Kiev in November-December 1240AD caused a demographic contraction of 53,2% from the 1200AD peak, reducing the regional population to 740k in 1250AD before partial rural and small-town recovery brought the Golden Horde *baskak*-administered appanages to 820k by 1300AD [10][11].

Background

Geographically, the region covers the Middle Dnieper forest-steppe, the southern Pripyat-Teterev-Uzh Polesian woodlands, the Upper Southern Bug basin and the Lower Dnieper Pontic Steppe corridor (~265.000 km^2 across modern central and southern Ukraine). Historically, this territory formed the metropolitan core of unified Kievan Rus (863-1133AD) and later divided into two Rurikid principalities and their steppe marches during the period of dynastic devolution (1133-1300AD).

The Principality of Kiev (1133-1322AD, ~95.000 km^2) occupied the Right-Bank Middle Dnieper upland plateau and the Irpin, Stuhna, Ros (*Porossia*), Teterev, Uzh and Sluch river basins (including the Drevlian Ovruch Volost), together with the agricultural buffer of the Bolokhov Land on the Upper Southern Bug and Horyn headwaters. Across the river, the Principality of Pereyaslavl (1133-1364AD, ~68.000 km^2) occupied the Left-Bank Middle Dnieper chernozem forest-steppe across the Trubezh, Supoy, Oster, Sula (*Posullya*), Khorol, Psyol, Vorskla and Upper Seversky Donets basins, serving as the eastern bulwark of Southern Rus against the Pecheneg, Tork and Polovtsian (Cuman) confederations. To the south, the Lower Dnieper Emporium and Pontic Steppe Frontier (~102.000 km^2) encompassed the Dnieper Rapids (*Porohy*), the Khortytsia-Protolche river crossing, the Kievan maritime enclave of Oleshye (46,62°N, 32,62°E) at the Dnieper-Bug estuary and the winter pastures (*zimovniki*) of the Black Sea steppe between the Lower Southern Bug and the Molochnaya-Azov coast.

Results

Prehistoric and Early Iron Age Settlement (10000BC - 500AD)

Upper Palaeolithic, Neolithic and Trypillia Mega-Sites (10000BC - 2000BC)

Human settlement in the Middle Dnieper valley during the terminal Pleistocene (10000BC) was anchored by Epigravettian mammoth-hunter base camps occupying periglacial loess river terraces along the Dnieper, Ros, Supoy and Udai valleys. Excavations at Mezhirich (near Kaniv), Dobranichevka, Gontsy (on the Udai) and Kirillovskaya (in the Podol district of Kiev) reveal circular mammoth-bone dwellings housing winter groups of 30 - 60 hunter-gatherers [1][12]. Across the Middle and Lower Dnieper basin, seasonal band territories support an estimated Upper Palaeolithic population of 14k (14.000) at 10000BC. Following the Mesolithic-Neolithic transition (6000BC), represented by the Bug-Dniester Culture on the Right Bank and the Dnieper-Donets Culture along the Dnieper Rapids (Nikolskoye, Vovnigi, Dereivka) and Left-Bank tributaries, mixed foraging, fishing and early stockbreeding communities raised regional population to 42k (42.000).

During the Middle and Late Eneolithic (4200BC - 3200BC), the forest-steppe interfluve between the Southern Bug, the Hnylyi Tikych and the Ros-Dnieper line became one of the most densely settled zones of Eneolithic Europe with the rise of the Cucuteni-Trypillia Culture (Stages BII and CI, peaking around 3600BC). Geomagnetic surveys and excavations directed by V. A. Kruts, M. Yu. Videiko and international teams have documented a cluster of concentric mega-sites (*gigant-poseleniia*) on the Cherkasy-Uman plateau that functioned as fortified agro-towns during periods of inter-group competition and steppe encroachment [1][2][13]. Talianky (48,8°N, 30,53°E, Stage CI, c. 3700-3500BC) covered 450 ha with ~2.700 geomagnetic house anomalies (n=2700) arranged in concentric ovals; accounting for synchronous house-use horizons (75-80% contemporaneous occupation at 5 - 7 persons per two-storey *ploshchadka* dwelling), Talianky housed 15k (15.000) inhabitants at 3600BC. Maidanetske (48,82°N, 30,69°E) enclosed 200 ha with ~2.000 - 2.300 structures arranged in four concentric rings and public assembly buildings (*mega-structures*), supporting 10k (10.000) inhabitants. Nebelivka (48,64°N, 30,56°E) formed a 260 ha Stage BII mega-site with a perimeter ditch, two-storey temple complex and ~1.450 dwellings (n=1450), supporting 9k (9.000) inhabitants. Companion nucleated centres stood at Dobrovody (48,77°N, 30,38°E, 250 ha, 8,5k), Sushkivka (120 ha, 4,5k), Vesely Kut (150 ha, 5k), Kosenivka (120 ha, 4k), Chichirkozovka (100 ha, 3,5k) and the eponymous Trypillia (Tripolye) complex on the Middle Dnieper bluffs (2,5k), alongside Late Trypillia CII Sofievka and Chapayevka cemeteries on the Kievan Dnieper terraces [1].

Combining the Uman-Cherkasy mega-site cluster (~65k in nucleated settlements >80 ha), the Right-Bank Middle Dnieper and Teterev agricultural villages (~145k) and Early Usatovo and Lower Dnieper pastoral communities (~70k), the regional population reached an Eneolithic peak of 280k (280.000) at 3600BC. With the aridification of the late fourth millennium BC and the expansion of Yamnaya (Pit-Grave), Catacomb and Middle Dnieper Corded Ware pastoralists by 2000BC, nucleated mega-sites gave way to mobile steppe kurgan communities and smaller riverside settlements (such as the copper-working and horse-breeding centre at Dereivka and fortified river bluffs at Pustynka), bringing regional population to 195k (195.000) in 2000BC.

Scythian Hillforts, Greek Poleis and the Chernyakhov Horizon (500BC - 500AD)

In the Early Iron Age (700BC - 300BC), trade in forest-steppe grain, hides, wax and slaves for Milesian Greek wine, olive oil and metalwork through the Lower Bug-Dnieper estuary turned the Middle Dnieper into the agricultural heartland of Herodotus's Scythian ploughmen (*Skolotoi*), Geloni and Budini [3][14]. This era produced two major settlement networks at the 500BC benchmark. Across the forest-steppe hillforts (*gorodishcha-giganty*) on the Left-Bank Vorskla watershed, the Belsk Hillfort (identified by B. A. Shramko with Herodotus's Gelonus, 50,09°N, 34,63°E) enclosed 4.020 ha within 33,8 km of earthen ramparts uniting the Western (72 ha), Eastern (65,2 ha) and Kuzeminsk (15,4 ha) citadels, housing bronze-casting, iron-smelting and Greek pottery workshops with an estimated population of 12k (12.000) [3]. On the Right Bank and Left-Bank Sula-Trubezh rivers stood a ring of large fortified centres:

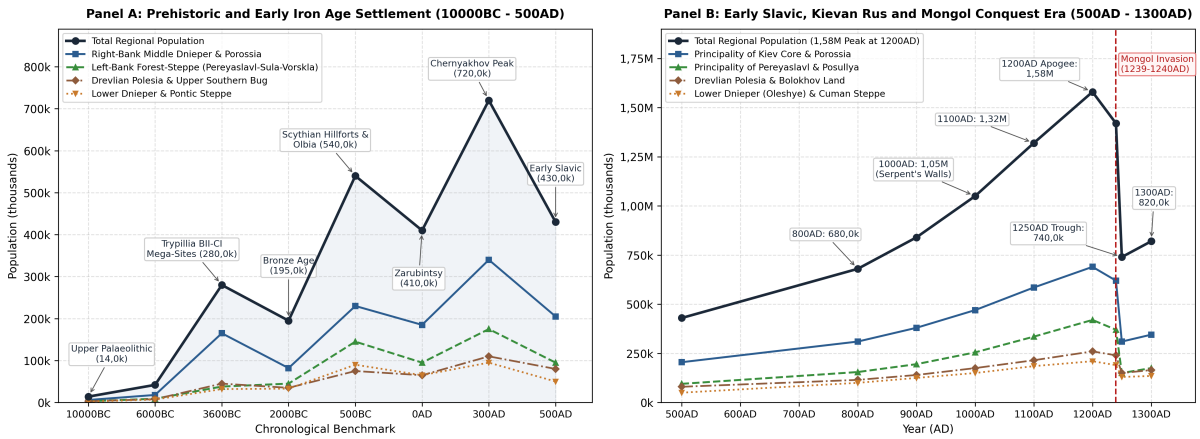


Figure 2. Population of the Middle Dnieper core across prehistoric, classical and medieval periods (10000BC - 1300AD).

Trakhtemyriv on the Kaniv Dnieper bend (500 ha, 4,5k), Khotiv on the southern outskirts of Kiev (31 ha, 2,5k), Motronyn in the Cherkasy Cold Ravine (200 ha, 4k), Pastyrskoye (25 ha, 2,5k), Nemirov on the Southern Bug (110 ha, 4k), Basovka on the Upper Sula (170 ha, 3k) and Knyazha Hora (1,8k) [14]. At the same time, along the Lower Bug-Dnieper coast, Milesian colonists in the mid-seventh to early sixth centuries BC founded Pontic Olbia (Parutino, 46,69°N, 31,9°E, 50 ha stone-walled upper and lower city plus a rural *chora* network of over 100 sites along the Berezan-Bug-Dnieper liman), which reached 15k (15.000) inhabitants in the urban core at 500BC, complemented by the island emporium of Berezan (Borysthenes, 2,5k), Nikonion (2k) and Glubokaya Pristan (1,2k) [4].

Together with dense open agricultural *selishcha* across the Kiev-Cherkasy plateau and royal Scythian pastoralist winter complexes in the Lower Dnieper (precursors to Kamenskoye Gorodishche, 1.200 ha, which peaked in the fourth century BC at 6k), the Middle and Lower Dnieper supported 540k (540.000) inhabitants at 500BC.

Following the Sarmatian disruption of the third and second centuries BC, the Zarubintsy Culture (200BC - 100AD, named after the cemetery near Trakhtemyriv and Pereyaslavl, with key settlements at Pilipenkova Hora at Kaniv, Chodosovka, Liutezh iron-smelting works north of Kiev and Subotiv) maintained a riverine population of 410k (410.000) at 0AD alongside the late Scythian stone fortresses of the Lower Dnieper (Zolota Balka, Gavrilovka, Lyubimovka, Znamenskoye) [15]. During the Late Roman period (250AD - 375AD), the multi-ethnic Chernyakhov Culture introduced heavy iron ploughshares, rotary querns and wheel-thrown grey-ware pottery across more than 1.200 documented unfortified valley settlements (n=1200) in the Kiev, Pereyaslavl and Cherkasy forest-steppe (Obukhiv, Zhukivtsi, Chernyakhiv, Bilozerka, Kompaniiets and Velikaya Bugaevka). Archaeological work by B. V. Magomedov estimates an average village size of 15 - 25 homesteads (90 - 150 inhabitants) spaced every 2 - 4 km along minor streams, bringing the regional population to a pre-medieval peak of 720k (720.000) at 300AD [5]. The Hunnic invasion of 375AD and the collapse of Pontic Roman trade networks caused a sharp contraction; by 500AD, the Early Slavic Prague-Korchak Culture (in the Right-Bank Polesian and Teterev woodlands) and the Penkovka Culture (associated with the *Antae* along the Ros, Tyasmin, Sula and Psyol) alongside the early hilltop settlement on Starokyivska Hora (Kiev, 1,5k at 500AD) and the metallurgical craft centre of Pastyrskoye (2,5k in the sixth and seventh centuries AD) sustained 430k (430.000) inhabitants [16][17].

Table 1. Prehistoric, antique and early Iron Age population estimates across the Middle Dnieper sub-regions (10000BC - 500AD).

Benchmark	Right-Bank Kiev & Porossia	Drevlian & Bolokhov Land	Left-Bank Pereyaslavl & Sula	Lower Dnieper & Steppe	Regional Total
10000BC	6k	3k	3k	2k	14k
6000BC	18k	8k	9k	7k	42k
3600BC	165k	45k	38k	32k	280k
2000BC	82k	35k	45k	33k	195k
500BC	230k	75k	145k	90k	540k
0AD	185k	65k	95k	65k	410k
300AD	340k	110k	175k	95k	720k
500AD	205k	80k	95k	50k	430k

Early Slavic and Unified Kievan Rus (500AD - 1133AD)

Luka-Raikovetska, Volyntsevo and the Rise of Kiev (800AD - 900AD)

During the eighth and ninth centuries AD (800AD benchmark), Eastern Slavic agrarian settlement expanded on both banks of the Middle Dnieper. On the Right Bank, the Luka-Raikovetska Culture established fortified tribal centres (*grady*) and rural clusters across the Polanian (*Poliane*), Drevlian (*Drevliane*) and Ulichian (*Ulichi*) lands, including Starokyivska Hora, Zamkova Hora and the earliest dendrochronologically dated log structures in Podol (887AD horizon at Kiev, 5k in 800AD, rising to 12k in 900AD) [6][18], Iskorosten (Korosten) on the Uzh (1,8k in 800AD), Malyn, Raiky and Monastyrek on the Kaniv Dnieper bend (1,2k). On the Left Bank, the Volyntsevo-Romny Culture of the Severians (*Severiane*) occupied the Desna-Seim-Sula-Psyol forest-steppe (Opishnia, Bititsa 2k, Romny or *Romen* 1,2k, Donets 1k), operating initially within the tributary orbit of the Khazar Khaganate before Oleg of Novgorod consolidated the Dnieper axis in 882AD [19][20]. Regional population recovered to 680k (680.000) at 800AD and reached 840k (840.000) at 900AD as Scandinavian-Slavic *druzhina* retinues organised the *polyudie* winter tribute circuit linking the Drevlian, Polanian and Severian river systems to the Byzantine trade treaties of 907AD and 911AD.

Vladimir the Great, Yaroslav the Wise and the Serpent's Walls (1000AD - 1100AD)

The conversion of Rus under Vladimir Svyatoslavich (980-1015AD) and the institutional consolidation under Yaroslav the Wise (1019-1054AD) reshaped the settlement landscape of the Middle Dnieper through state-directed frontier engineering and military colonisation. As recorded in the *Primary Chronicle* (*Povest' vremennykh let*, sub anno 988AD and 991AD), Vladimir declared that it was not good that there were so few towns around Kiev, and built concentric defensive lines along the Stuhna, Irpin, Trubezh, Oster, Ros and Sula rivers, staffing them with warriors relocated from the Slovenes of Novgorod, the Krivichi, the Chud and the Vyatichi [21][22].

Archaeological excavations by M. P. Kuchera across the Serpent's Walls (*Zmiyevy Valy*) show that between the late tenth and mid-eleventh centuries, the Kievan state built over 950 km of continuous timber-laced earthen ramparts (10 - 12 m high, fronted by 5 - 6 m ditches) arranged in six echelons: the Vita-Bobrytsia, Stuhna, Irpin-Teterev, Ros (Porossia), Pereyaslavl-Trubezh and Sula (Posullya) lines [8]. Shielded behind

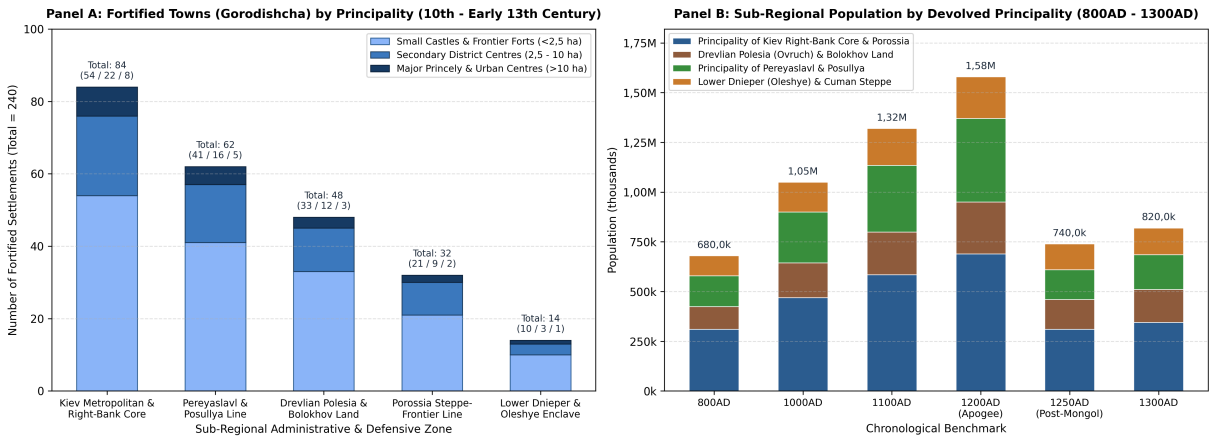


Figure 3. Settlement hierarchy and principality population breakdown across the Middle Dnieper core (800AD - 1300AD).

these ramparts, rural *selishcha* density tripled between 950AD and 1100AD. At the 1000AD benchmark (1,05M regional total), Kiev expanded from Vladimir's 10 ha *Detinets* (hosting the Church of the Tithes, consecrated 996AD) and the artisans' quarter of *Podol* to 28k (28.000) inhabitants, surrounded by the fortified royal residences and garrison towns of Belgorod (110 ha enclosure founded 991AD, 6k), Vyshhorod (70 ha, 5k), Vasiliev (Vasylkiv, 2,5k), Trepol (Trypillia, 2k), Vitichev (1,8k) and the Left-Bank capital of Pereyaslavl-Russky (80 ha fortified complex built 992AD, 8k) [6][23]. By the 1100AD benchmark (1,32M regional total), under Yaroslav the Wise (who built the 70 ha *City of Yaroslav* enclosed by 3,5 km of 16 m-high ramparts, the Golden Gate and St Sophia Cathedral in 1037AD, and founded the Porossia frontier line along the Ros in 1032AD around Yuriev or Bila Tserkva, Torchesk and Korsun) and Vladimir Monomakh (who led the anti-Polovtsian coalitions from Pereyaslavl after the Congresses of Lyubech in 1097AD and Dolobsk in 1103AD), Kiev reached 40k (40.000) inhabitants and Pereyaslavl-Russky reached 12k (12.000) [9][24].

Devolved Principalities (1133AD - 1240AD)

Following the death of Mstislav the Great (1132AD), the unified realm divided into autonomous dynastic lands. At the 1200AD benchmark, representing the demographic peak of pre-Mongol Southern Rus, the Middle and Lower Dnieper region reached 1.580.000 (1,58M) inhabitants across 240 excavated fortified settlements (*gorodishcha*, n=240) catalogued by A. V. Kuza [7][23] and O. P. Motsia [9] and over 2.800 open rural villages (*selishcha*, n=2800).

The Kievan Metropolis (45k - 50k at 1200AD)

Planimetric and stratigraphic studies by P. P. Tolochko [6][12][24], M. A. Sahaydak [18] and H. Yu. Ivakin [10] show that late twelfth- and early thirteenth-century Kiev formed a multi-part urban complex spanning 360 - 400 ha (380 ha standard benchmark). The Upper City (*Gora*, ~98 ha) consisted of *Vladimir's City* (10-12 ha, princely *dvor* chambers, stone churches and boyar compounds), *Yaroslav's City* (70 ha, metropolitan cathedral of St Sophia, St Irene, St George and dense residential quarters), *Izyaslav-Svyatopolk's City* (8 ha, St Michael's Golden-Domed Monastery) and *Kopyrev Konets* (8 ha on the north-western spur). Below the bluffs, the Lower City (*Podol*, 180 - 200 ha) has been investigated through dendrochronological excavations across *Kontraktova Ploshcha*, *Zhytniy Rynok*, *Verkhniy Val* and *Nyzhniy*

Val, revealing standardised 300 - 600 m^2 fenced urban homesteads (*usadby*) containing 2 - 3 log dwellings, craft workshops (glass-making, bone-carving, enamelling, shoemaking, smithing) and merchant warehouses (*pirohy*). Allowing 3.500 - 4.000 active homesteads across *Podol* at 6 persons per household yields 22k - 25k residents in the Lower City alone [18]. Surrounding the core lay suburban monastic and princely satellites (~80 - 100 ha), including the *Kievo-Pecherskaya Lavra* (Caves Monastery), the princely residence of *Berestovo*, *Klov*, *Dorogozhichi*, *Vydubychi*, *Kyrylivska*, *Kitaevo* (35 ha fortified complex on the southern Dnieper hills) and *Gnilets*.

Combining Upper City boyar and clerical households (~15k), *Podol* artisan and merchant homesteads (~24k), *Kopyrev Konets* and the suburban districts (~9k - 11k), Tolochko's estimate of 50k (50.000) inhabitants at 1200AD (equivalent to ~131,6 persons/ha across 380 ha) places Kiev among the largest cities of twelfth-century Europe alongside Constantinople, Paris and London [24][25].

The Right-Bank Kievan Ring and Stuhna-Dnieper Corridor

Protecting the southern and western approaches to Kiev stood a network of princely seats, custom ports and fortress-towns (84 fortified *gorodishcha* in the Kiev Core zone: 8 major >10 ha, 22 secondary 2,5-10 ha and 54 small <2,5 ha castles). Belgorod (Bilhorodka) on the Irpin (50,39°N, 30,23°E, 110 ha *detinets* and *okolny gorod*, episcopal see and secondary capital of Rurik Rostislavich) supported 9k (9.000) inhabitants at 1200AD. Vyshhorod (50,58°N, 30,49°E, 70 ha cliffside citadel and potter's *posad*, shrine of SS Boris and Gleb) housed 7k (7.000), while Kaniv (49,75°N, 31,47°E, 24 ha, Dnieper crossing and assembly point against the Cumans, with the stone Church of St George built in 1144AD) reached 4k (4.000). Additional fortified centres included Vasiliev (Vasylkiv) on the Stuhna (20 ha, 3,5k), Trepol (Trypillia, 18 ha Dnieper cliff fortress controlling the Stuhna mouth, 3k), Roden on *Knyazha Hora* (16 ha craft town near Kaniv, 2,5k), Vitichev (14 ha beacon fortress on the Dnieper, 2,2k), Zarub (1,5k), Chuchin (12 ha, 1,8k), Khalep (1,4k), Svyatoslavl, Ivan-Gorod (1,5k), Tumasch (1,4k), Kitaigorod (1,5k) and Zvenyhorod-Kievan (1,6k) [7][9].

The Porossia Steppe-Frontier Line

Along the Ros River valley (80 km south of Kiev), the Kievan grand princes settled semi-nomadic Turkic federates, the Chorni Klobuky ("Black Hoods", uniting the Torks, Berendei, Kovui, Turpei and remnant Pechenegs), as hereditary border cavalry endowed with fortified towns (32 *gorodishcha* in the Porossia zone: 2 major, 9 secondary, 21 small) [9][22]. Torchesk (49,68°N, 30,63°E, identified with the 45 ha *Olshanitsa* or *Sharki* complex on the Gorokhovatka-Ros) served as the capital of the Black Hoods and appanage seat of Mstislav the Bold (1193-1207AD), supporting 6k (6.000) inhabitants at 1200AD. Yuriev (Bila Tserkva, 49,79°N, 30,11°E, 28 ha *Zamkova Hora* and *posad*, episcopal seat of the Bishop of Yuriev) housed 4,5k (4.500), flanked by Korsun on the Ros (49,42°N, 31,26°E, 16 ha, 2,8k), Boguslavl (49,55°N, 30,87°E, 15 ha, 2,5k), Volodarev (Volodarka, 1,8k), Rostovets (1,8k), Dveren (1,5k), Polksten (1,5k), Bakhmutov (1,2k), Sakkov (1,1k) and Borovyi (1k). Together, the Right-Bank Kiev Core and Porossia frontier supported 690k (690.000) inhabitants at 1200AD.

Drevlian Polesia and the Bolokhov Land (~260k at 1200AD)

North-west and south-west of the Kievan core lay two sub-regions encompassing 48 fortified *gorodishcha* (3 major >10 ha, 12 secondary 2,5-10 ha, 33 small <2,5 ha). In Drevlian Polesia, settlement was centred on Ovruch (Vruchiy, 51,32°N, 28,8°E, 25 ha, brick Church of St Basil built by Rurik Rostislavich c. 1190AD, 4,5k at 1200AD) and Iskorosten (Korosten, 50,95°N, 28,64°E, four-citadel complex on the Uzh, 22 ha, 4k). As shown by A. P. Tomashevsky [26], the Ovruch ridge was the sole European source of pink pyrophyllite slate, surrounded by over 40 specialised workshop villages (n=40; *Nagorany*, *Kamenka*, *Korenevka*) manufacturing slate spindle-whorls (*priaslitsa*) and sarcophagi exported across Rus, Poland and Volga Bulgaria. Secondary Drevlian centres included Vozvyagl (Zviahel) on the Sluch (16 ha, 2,8k), Malyn (1,8k),

Kotelnich (2k), Zhytomyr (1,8k), Horodsk (*Gorodesk*, 2k), Michesk (Radomyshl, 1,6k), Ushchesk (Ushomyr, 1,5k) and Chornobyl (1,5k at the Pripyat-Uzh confluence). To the south-west, the Bolokhov Land (*Bolokhovskaia Zemlia*, 1150-1257AD) occupied the fertile forest-steppe plateau of the Upper Southern Bug, Sluch and Horyn between the principalities of Kiev, Galicia-Volhynia and the Cuman steppe; combining the Right-Bank core (690k) with the Drevlian and Bolokhov lands (260k) gives 950k across the Principality of Kiev at 1200AD. Governed by autonomous local elites (*knyaztsi bolokhovskie*), its fortified towns, including Bolokhov (Lyubar, 15 ha, 2,5k), Gubin (12 ha, 2k), Kudin (1,8k), Kobud (1,6k), Derevich (1,6k), Bozhsky (*Bozhesk*, 2k), Mezhibozhye (Medzhybizh, 2,2k) and Dyadkov (1,4k), were known for wheat and millet cultivation [9][27].

Principality of Pereyaslavl (~420k at 1200AD)

East of the Dnieper, the Principality of Pereyaslavl served from the eleventh to early thirteenth centuries as the ancestral patrimony of the Monomakhovichi and the main military shield of Southern Rus [19][28]. Archaeological surveys by Yu. Yu. Morgunov and O. V. Sukhobokov record 62 fortified *gorodishcha* (n=62) across the Pereyaslavl land (5 major >10 ha, 16 secondary 2,5-10 ha, 41 small <2,5 ha). Located at the confluence of the Alta and Trubezh rivers, the capital of Pereyaslavl-Russky (Pereiaslav, 50,07°N, 31,46°E) comprised a 10 ha *Detinets* (containing Bishop Ephraim's stone Cathedral of St Michael built in 1089AD, stone episcopal gates, bathhouse and princely palace) and a 70 ha fortified *Okolny Gorod* plus unfortified riverside *posady* (80 ha walled core, ~100 ha total complex), supporting 15k (15.000) inhabitants at 1200AD [6][7].

Stretching 180 km from the Dnieper estuary of the Sula to the Upper Sula border with the Kursk-Putivl Severians, the Posullya defensive line comprised 18 inter-visible timber-and-earth *gorodishcha* (n=18) anchored at the mouth by the fortified harbour-citadel of Voin (Voinska Hreblia, 18 ha with an artificial dredged interior ship-harbour, 2,5k at 1200AD), followed upstream by Zheldi (1,4k), Rimiv (14 ha, besieged by Konchak in 1185AD, 2,2k), Lukoml (13 ha, 2k), Lubno (Lubny, 16 ha, founded by Monomakh, 3k), Khorol (2k), Sneporod (12 ha, 1,8k), Ksniatin (13 ha, 2k), Sinets (1,5k), Romen (Romny, 15 ha, 2,5k) and Glinsk (1,5k) [28]. Behind the Sula line in the interior Trubezh-Supoy-Oster basins lay Priluk (Pryluky) on the Udai (15 ha, 2,8k), Nosivka (2,2k), Oster (Gorodets-Ostersky, 14 ha, founded by Monomakh in 1098AD with St Michael's *Yurievska Bozhnytsia*, 2,5k), Baruch (1,8k), Bronnitsa (1,6k), Pesochen (1,5k) and Zholnin (1,4k), while the south-eastern forest-steppe corridor along the Psyol, Vorskla and Seversky Donets hosted Ltava (Poltava, 49,58°N, 34,56°E, 14 ha on *Ivanova Hora*, mentioned 1174AD, 2,2k), Goltav (1,5k), Nizhegolsk (1,4k) and the easternmost frontier fortress of Donets (49,85°N, 36,21°E, Khorosheve hillfort south of modern Kharkiv, where Igor Svyatoslavich escaped from Polovtsian captivity in 1185AD, 2k) [19][29].

Lower Dnieper Emporium and the Steppe Frontier (~210k at 1200AD)

South of the Porossia and Posullya ramparts extended the Pontic Steppe (*Pole Polovetskoe*), where 14 fortified river-island and delta trading stations (n=14) coexisted with the seasonal camps (*vezhi* and winter *zimovniki*) of the Dnieper Polovtsian confederations (the Burchevichi and Lukomorie hordes led by Khans Kobyak, Bonyak and Kotyan) [9]. At Oleshye (46,62°N, 32,62°E, on Veliky Potemkin Island opposite modern Kherson and Oleshky), the maritime gateway of Kievan Rus occupied a 20 ha waterlogged port settlement yielding thousands of Byzantine amphorae, lead seals of Kievan princes and Greek officials, glass bracelets and ship fittings, where Kievan military flotillas met Byzantine merchants and newly consecrated Metropolitans, supporting 3,5k (3.500) permanent inhabitants at 1200AD. Upstream at Khortytsia (Protolche, 47,82°N, 35,07°E), the assembly point below the Dnieper Rapids (*Khortytskyi Ostrov* and *Protolche* ford), where the Rus princes mustered before the 1103AD Suten and 1223AD Kalka campaigns, supported 1,8k inhabitants at 1200AD, alongside Berestovo (Beloberezhye, 1,2k) at the Kinburn spit and Cuman winter towns along the Samara, Orel and Molochnaya rivers (210k total

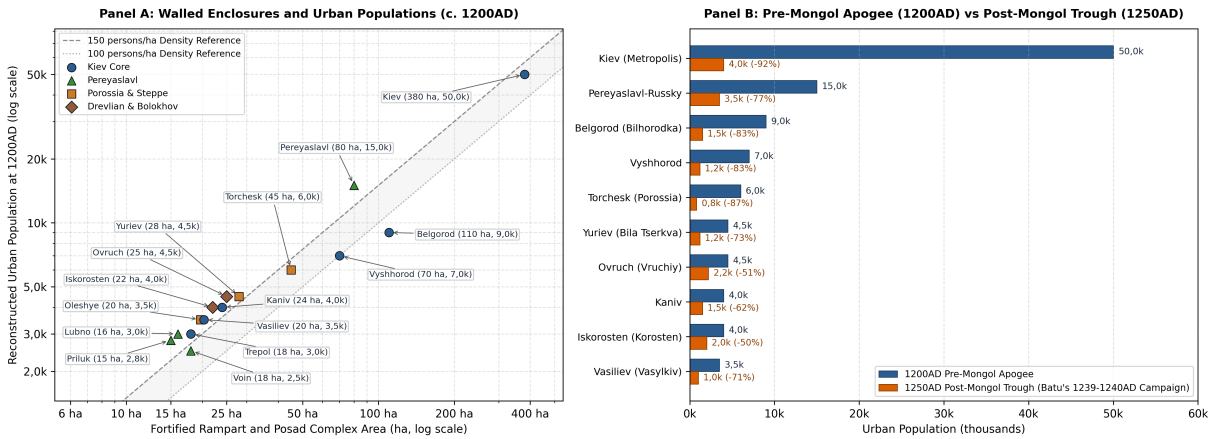


Figure 4. Urban archaeological hierarchy and demographic impact of the 1239-1240AD Mongol sieges in Southern Rus (1200AD - 1300AD).

sub-regional population at 1200AD).

Mongol Invasion and Post-Siege Recovery (1239AD - 1300AD)

Batu Khan's Campaigns of 1239-1240AD

Following the reconnaissance raid of Jebe and Subutai and the defeat of the Kievan, Galician, Chernigov and Smolensk princes at the Battle of the Kalka River (31 May 1223AD), where Grand Prince Mstislav Romanovich of Kiev and 10.000 Kievan warriors fell, the Middle Dnieper principalities entered the 1240AD benchmark with a slightly reduced population of 1,42M (1.420.000). The Mongol invasion of Southern Rus unfolded in two winter-spring campaigns [6][10][29].

In March 1239AD, Mongol armies under Möngke and Batu breached the Posullya ramparts, burning Rimiv, Voin, Lubno, Lukoml and Priluk, and stormed Pereyaslavl-Russky on 3 March 1239AD. As recorded in the *Laurentian* and *Hypatian Chronicles*, St Michael's Cathedral was demolished, Bishop Simeon was killed and the city's population was slain or led into captivity. Archaeological excavation across the Pereyaslavl *Detinets* and *Posullya* hillforts reveals a continuous 1239AD burn layer containing thousands of Mongol-type rhomboid armour-piercing arrowheads, unburied skeletons in burned cellars and buried treasure hoards (*klady*) that were never recovered [28]. After Möngke's reconnaissance of Kiev from *Pesochen (Gorodok)* on the Left Bank in late 1239AD, Batu Khan's main army crossed the Dnieper in autumn 1240AD, overrunning the Porossia federate towns (Torchesk, Yuriev, Korsun, Vitichev, Vasiliev, Belgorod and Vyshhorod). Defended by Daniel of Galicia's *voivode* Dmitr, Kiev was surrounded by Mongol siege engines (*poroki*), which breached the Lyadskie Gates of Yaroslav's City before a final stand around the Church of the Tithes, which collapsed under the weight of refugees on 6 December 1240AD.

Mass graves excavated by M. K. Karger, P. P. Tolochko and H. Yu. Ivakin in the Kiev Upper City (over 2.000 skeletons in sanitary pits near the Church of the Tithes and St Sophia, alongside burned workshops with slain artisans inside) attest to the scale of the urban destruction [6][10]. When the Papal envoy Giovanni da Pian del Carpine (John of Plano Carpini) passed through the Kievan land and visited Kiev in February 1246AD, he described the post-conquest landscape:

"They attacked Rus, where they made great havoc, destroying cities and fortresses and slaughtering

> *men; and they laid siege to Kiev, the capital of Rus; after they had besieged the city for a long time, they took it and put the inhabitants to death. When we were journeying through that land we came across countless skulls and bones of dead men lying about on the ground. Kiev had been a very large and densely populated town, but now it has been reduced almost to nothing, for there are at the present time scarce two hundred houses there and the inhabitants are kept in utter slavery."* [11]

Allowing for Carpini's ~200 taxable household compounds (*dvory*) on *Podol* plus surviving clerical and suburban households around the Caves Monastery and *Kopyrev Konets*, H. Yu. Ivakin estimates Kiev's population at 3,5k - 4,5k (4k benchmark) in 1250AD, a 92% drop from its 1200AD peak of 50k [10].

The Bolokhov Land and Late Thirteenth-Century Recovery (1250AD - 1300AD)

Archaeological and chronicle evidence shows that population loss during the 1239-1240AD invasion was spatially uneven across the Middle Dnieper sub-regions. Heavy depopulation occurred in the exposed forest-steppe (-64% in Pereyaslavl and Posullya; -55% in Kiev Core and Porossia): along the Sula and Ros frontier lines, nearly 80% of fortified *gorodishcha* (Voin, Rimiv, Sneporod, Torchesk, Roden, Vitichev) were permanently abandoned after 1239-1240AD [7][28]. At the 1250AD post-conquest trough, the total regional population stood at 740k (740.000): 310k in the Right-Bank Kiev Core and Porossia, 150k in the Drevlian and Bolokhov lands, 150k in the Principality of Pereyaslavl and 130k in the Lower Dnieper and steppe zone.

Greater resilience prevailed in forested Drevlian Polesia and the Bolokhov Land. The marshy woodlands of the Teterev-Uzh-Pripyat basin (Ovruch, Iskorosten, Chornobyl, Zhytomyr) sheltered rural refugees fleeing the Dnieper corridor. At the same time, the princes and towns of the Bolokhov Land (Bolokhov, Gubin, Kudin, Kobud, Bozhsky) escaped destruction in 1240AD by submitting directly to the Mongols as *Tataropoli* ("Tatar-tribute ploughmen"), sowing wheat and millet for Batu's troops until King Daniel Romanovich of Galicia-Volhynia launched punitive expeditions in 1241AD and 1255-1257AD that razed the Bolokhov ramparts and Vozvyagl [9][27].

Following the Golden Horde censuses (*chislo*) of 1257-1259AD and 1273AD under Mengu-Timur, the Principalities of Kiev and Pereyaslavl were administered by local Rurikid and Putivl-Olgovichi princes alongside Golden Horde *baskaki* (governors and tax-collectors). Although Metropolitan Maximus transferred the residence of the Metropolitanate of All Rus from Kiev to Vladimir-on-the-Klyazma in 1299AD, Ivakin's excavations show steady late thirteenth-century craft and commercial revival in Kievan *Podol*, *Pechersk* and the castle on *Zamkova Hora* (8,5k inhabitants at 1300AD), Pereyaslavl (4,5k), Ovruch (3k), Iskorosten (2,6k), Kaniv (2,2k) and Yuriev (2k), bringing the regional population to 820k (820.000) at 1300AD before the Lithuanian conquest under Gediminas in the 1320s AD [10].

Table 2. Medieval sub-regional population across the principalities of Kiev and Pereyaslavl (500AD - 1300AD).

Benchmark	Kiev Core & Porossia	Drevlian & Bolokhov Land	Pereyaslavl & Posullya	Lower Dnieper & Steppe	Regional Total
500AD	205k	80k	95k	50k	430k
800AD	310k	115k	155k	100k	680k
900AD	380k	140k	195k	125k	840k
1000AD	470k	175k	255k	150k	1,05M (1,05M)
1100AD	585k	215k	335k	185k	1,32M (1,32M)
1200AD	690k	260k	420k	210k	1,58M (1,58M)
1240AD	620k	240k	370k	190k	1,42M (1,42M)
1250AD	310k	150k	150k	130k	740k
1300AD	345k	165k	175k	135k	820k

Cross-Regional Comparison (1000AD - 1300AD)

To maintain consistency across the historical dataset, the table below compares the Middle Dnieper Core (Kiev and Pereyaslavl) with the northern and eastern Rus principalities documented in the companion European Russia regional review [21][23][30].

Table 3. Cross-regional population comparison between the Middle Dnieper core and northern and eastern Rus (1000AD - 1300AD).

Polity/Regional Unit	1000AD Pop.	1100AD Pop.	1200AD Apogee	1250AD Trough	1300AD Pop.
Principality of Kiev (incl. Drevlian & Bolokhov)	645k	800k	950k	460k	510k
Principality of Pereyaslavl (incl. Posullya)	255k	335k	420k	150k	175k
Lower Dnieper (Oleshye) & Pontic Steppe Frontier	150k	185k	210k	130k	135k
Kiev and Pereyaslavl Subtotal	1,05M	1,32M	1,58M	740k	820k
cf. Vladimir-Suzdal/Rostov (European Russia)	340k	510k	780k	520k	640k
cf. Novgorod & Pskov Republics (European Russia)	290k	380k	480k	460k	510k
cf. Principality of Smolensk (European Russia)	210k	275k	340k	290k	310k
cf. Kursk-Rylsk & Bryansk-Upper Oka (European Russia)	280k	390k	500k	270k	310k
cf. Total European Russia Benchmark	2,15M	2,9M	3,85M	2,8M	3,1M

Conclusion

This comparison illustrates the demographic shift of the twelfth and thirteenth centuries: while the Middle Dnieper Core (1,58M at 1200AD) remained the most densely populated and urbanised territory of the Rus world up to the eve of the Mongol invasion, containing two of the five largest cities (Kiev 50k and Pereyaslavl 15k) and 240 fortified *gorodishcha* (n=240), its open forest-steppe geography left it exposed to the 1239-1240AD Mongol siege campaigns and subsequent nomadic raids. Whereas the total population of European Russia contracted by 27,3% (3,85M to 2,8M) between 1200AD and 1250AD (buffered by the uninvaded Novgorod-Pskov and Smolensk lands) and recovered to 80,5% of its pre-Mongol peak by 1300AD (3,1M), the Middle Dnieper Core suffered a 53,2% contraction (1,58M to 740k in 1250AD) and stood at 51,9% of its pre-Mongol peak (820k) at the 1300AD benchmark.

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